

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Engineered **inference pipelines** for AI systems, reducing average response latency by **30%** through kernel optimizations and efficient model serving strategies.
- Designed and deployed **model serving architectures** for diverse environments, including resource-constrained devices, achieving a **25%** reduction in memory footprint.
- Developed and monitored controlled **inference tests** in production, tracking key performance indicators like latency, throughput, and memory consumption to validate benchmark performance.
- Analyzed computational efficiency and diagnosed bottlenecks in serving pipelines, resolving suboptimal batch processing and high memory usage issues to improve scalability.
- Integrated optimized serving and inference frameworks into production pipelines for edge and on-device applications, achieving a **99.5%** success rate in real-world performance metrics.
- Prepared high-quality test datasets and simulation scenarios to effectively evaluate model performance, latency, and memory utilization under various operational conditions on low-resource devices.

SKILLS

AI & Machine Learning – Inference Optimization, Model Serving, Kernel Optimization, Compute Shaders, Metal Shading Language (MSL), Performance Metrics, Latency Reduction, Throughput Improvement, Memory Footprint Optimization.

Programming Languages – Python, Java, C++, R.

Backend Development – Java Spring Boot, REST APIs, FastAPI, Flask.

Data Analytics & BI – Power BI, DAX, Tableau, Power Query, Data Modelling, Data Warehousing, Statistical Analysis, R / RStudio.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux.

Databases – PostgreSQL, MySQL, Oracle, SQL.

Enterprise Systems – ERP Integration Support, Change Management, Transaction Monitoring, Reconciliation Workflows.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Edge AI Inference Optimization Framework

- Developed a custom inference engine using **Metal Shading Language (MSL)** and **Python** to optimize model execution on resource-constrained edge devices, achieving a **40%** reduction in latency.

- Engineered a scalable serving architecture that supported high-throughput processing of multi-modal AI models (text, image) with a minimized memory footprint of **150MB** per instance.
- Implemented comprehensive performance monitoring and alerting using **CloudWatch** and custom metrics, ensuring **99.9%** uptime and rapid identification of computational bottlenecks.

Real-time Payments Transaction Monitoring Dashboard

- Built a **Power BI** dashboard to visualize real-time transaction flows and operational alerts for a financial services client, improving issue detection by **35%**.
- Automated data ingestion and transformation using **Python** scripts and **SQL** queries against PostgreSQL, reducing manual reporting effort by **20 hours** per week.
- Collaborated with business analysts to define key performance indicators and operational metrics, ensuring alignment with client requirements for transaction monitoring and reconciliation workflows.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)